

HDOC – Experiment 11

Question 1

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations:

- 1. Find the sum of squares of its digits using a while loop.**
- 2. Find the sum of cubes of its digits using a for loop.**
- 3. Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop.**

❖ Understand the problem

Input

- Positive Integer
- User's Menu Choice

Output

- Sum of squares of digits
- Sum of cubes of digits
- Difference (Sum of Cubes – Sum of Squares)

❖ Test case table

Test case	Input (choice, number)	Expected output	Actual output	Result
1	1, 123	Sum of squares= 14	14	Pass
2	2, 123	Sum of cubes= 36	36	Pass
3	3, 123	Difference= 22	22	Pass
4	1, 456	Sum of squares= 77	77	Pass
5	2, 456	Sum of cubes= 405	405	Pass

❖ Algorithm

1. Start.
2. Display the menu with three options.
3. Read the user's choice.
4. Read the positive integer.
5. Store the original number in a temporary variable.
6. According to the menu choice:
 - Case 1: Use a while loop to calculate the sum of the squares of the digits.
 - Case 2: Use a for loop to calculate the sum of the cubes of the digits.
 - Case 3: Use a do-while loop to calculate both the sum of squares and the sum of cubes, then find their difference.
7. Display the result.
8. If the menu choice is invalid, display an error message.
9. Stop.

❖ Program in C

```
#include <stdio.h>

int main()
{
    int choice;

    int number, temp, digit;

    int sumSquare = 0, sumCube = 0, difference = 0;

    printf("==== MENU ==== \n");
    printf("1. Sum of Squares of Digits (While Loop) \n");
    printf("2. Sum of Cubes of Digits (For Loop) \n");
    printf("3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop) \n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &number);

    switch(choice)
    {
        case 1:

            temp = number;

            sumSquare = 0;
```

```
while(temp > 0)
{
    digit = temp % 10;
    sumSquare += digit * digit;
    temp = temp / 10;
}

printf("\nSum of Squares of Digits = %d\n", sumSquare);
break;
```

case 2:

```
sumCube = 0;

for(temp = number; temp > 0; temp = temp / 10)
{
    digit = temp % 10;
    sumCube += digit * digit * digit;
}

printf("\nSum of Cubes of Digits = %d\n", sumCube);
break;
```

case 3:

```
temp = number;
sumSquare = 0;
sumCube = 0;
```

```
do
{
    digit = temp % 10;

    sumSquare += digit * digit;

    sumCube += digit * digit * digit;

    temp = temp / 10;

}while(temp > 0);

difference = sumCube - sumSquare;

printf("\nDifference = %d\n", difference);

break;

default:

    printf("\nInvalid Choice!\n");
}

return 0;
}
```

Compile, Execute, Evaluate

Test case 1

```
(kali㉿kali)-[~/Desktop/exp11]
$ ./switch
===== MENU =====
1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)
Enter your choice: 1
Enter a positive integer: 123

Sum of Squares of Digits = 14
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp11]
$ ./switch
===== MENU =====
1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)
Enter your choice: 2
Enter a positive integer: 123

Sum of Cubes of Digits = 36
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp11]
$ ./switch
===== MENU =====
1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)
Enter your choice: 3
Enter a positive integer: 123

Difference = 22
```

Test case 4

```
(kali㉿kali)-[~/Desktop/exp11]
$ ./switch
===== MENU =====
1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)
Enter your choice: 1
Enter a positive integer: 456

Sum of Squares of Digits = 77
```

Test case 5

```
(kali㉿kali)-[~/Desktop/exp11]
$ ./switch
===== MENU =====
1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)
Enter your choice: 2
Enter a positive integer: 456

Sum of Cubes of Digits = 405
```

❖ Compilation and execution screenshot

```
(kali㉿kali)-[~/Desktop/exp11]
$ vi switch.c

(kali㉿kali)-[~/Desktop/exp11]
$ gcc switch.c -o switch

(kali㉿kali)-[~/Desktop/exp11]
$ ./switch
===== MENU =====
1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)
Enter your choice: 1
Enter a positive integer: 123

Sum of Squares of Digits = 14

(kali㉿kali)-[~/Desktop/exp11]
$ ./switch
===== MENU =====
1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)
Enter your choice: 2
Enter a positive integer: 123

Sum of Cubes of Digits = 36

(kali㉿kali)-[~/Desktop/exp11]
$ ./switch
===== MENU =====
1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)
Enter your choice: 3
Enter a positive integer: 123

Difference = 22
```



```
(kali㉿kali)-[~/Desktop/exp11]
```

```
$ ./switch
```

```
===== MENU =====
```

1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)

Enter your choice: 1

Enter a positive integer: 456

Sum of Squares of Digits = 77

```
(kali㉿kali)-[~/Desktop/exp11]
```

```
$ ./switch
```

```
===== MENU =====
```

1. Sum of Squares of Digits (While Loop)
2. Sum of Cubes of Digits (For Loop)
3. Difference between Sum of Cubes and Sum of Squares (Do-While Loop)

Enter your choice: 2

Enter a positive integer: 456

Sum of Cubes of Digits = 405